

Question Paper

Exam Date & Time: 24-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Computer Networks [IT_3127]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Assume 1500-byte packet is transmitted from a host to a destination over a 100 Mbps wired link covering a distance of 1,200 km. The propagation speed in the medium is 2.5×10^8 m/s, and the processing delay at the destination is 2 ms. Assuming no queuing delay, calculate the total end-to-end delay experienced by the packet. (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) When a user sends an email over the Internet, multiple protocols and layers work together to ensure reliable communication. Make use of OSI model, explain how the functions of the Transport, Network, and Data Link layers interact during this process. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) A data packet of size 1200 bytes is transmitted in 4 milliseconds. Examine and calculate the transmission rate (bandwidth) of the link. (2)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2) A webpage consists of: (5)

- A)
- 1 HTML file of size 150 KB
 - 6 images each of size 350 KB
 - 2 CSS files of size 100 KB each

The network parameters are as follows:

- RTT = 80 ms
- Link bandwidth = 8 Mb/s

Each object (HTML, CSS, image) is requested separately over non-persistent HTTP. Examine the total time required to load the complete webpage.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Apply your understanding of email retrieval protocols to determine which protocol-POP3 or IMAP4- is more suitable when a user accesses their email from multiple devices such as a laptop and a smartphone. Justify your choice and explain how the message retrieval and storage mechanisms differ between the two protocols. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Explain how the HTTP protocol enables communication between a web browser (client) and a web server when a user enters a website URL. Describe the key steps involved in fetching and displaying the requested web page. (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 3) Assume the following dump of a UDP header in hexadecimal format: (5)
C312 0050 0024 9AF0

A) Answer the following:

- What is the source port number?
- What is the destination port number?
- What is the total length of the user datagram?
- What is the length of the data?
- Is the packet directed from a client to a server or vice versa?

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Apply your understanding of Selective Repeat ARQ system where with a sender window size of 4, the sender transmits packets 0, 1, 2, and 3. If packet 1 is lost but packets 2 and 3 are received correctly, explain how the receiver and sender handle this situation. How does Selective Repeat ensure reliability without retransmitting all packets? (3)

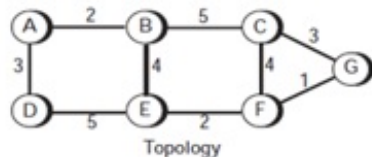
Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Make use of TCP data transfer, the sender transmits a segment with Sequence Number 3000, but the receiver never receives it due to network loss. Instead, it receives the next segment with Sequence Number 3500. Explain how the receiver responds in this situation and how the sender detects the lost segment to ensure reliable delivery. (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) (5)

A)



Analyze the given network topology, use Dijkstra's algorithm to determine the shortest path tree for router A. Show the sequence in which nodes are added to the path and state the shortest distance from A to all other routers (B, C, D, E, F, and G).

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Apply your understanding of IPv4 addressing and subnetting to determine the total number of addresses, the first address, and the last address in the network block that contains the IP address 73.22.17.25. Show the steps involved in your calculation. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) In classful IP addressing, show how are the Network ID and Host ID portions divided for different address classes (A, B, C, D, and E)? Explain conceptually which part identifies the network and which part identifies the host in each class. (2)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5) Assume the message sender wants to send is 1010001101, and the generator polynomial is $x^5 + x^4 + x^2 + 1$. Find the message transmitted by the sender. If the receiver receives the message, check if the receiver receives the correct message or not. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Analyse a given 4-bit data word that is transmitted using a parity bit for error detection. The 4-bit data word is 1011, and the parity bit used is even parity. After transmission, the received data is 1010. (3)

1. Was there an error during transmission?
2. If so, can the error be corrected using the parity bit? Why or why not?

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Explain how CSMA and CSMA/CD protocols help in minimizing collisions during data transmission. (2)

Course Outcome: "CO5" / Cognitive Level: "Understand"

-----End-----

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 IT_3127:Computer Networks

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 IT_3127:Computer Networks

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	6(12%)
3	Application	5	14(28%)
4	Analysis	7	30(60%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)